Course Notes

Abstract Algebra III

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Contents

Chapter 1

Lecture 1: Modules

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1.1 Basic Definitions and Examples

Definition 1.1.1. *Let R be a ring with 1, and let M be an abelian group. We say that M is a left R-module if R "left acts" on M:*

$$R \times M \to M, \quad (r, m) \mapsto rm$$

such that

$$\left.\begin{aligned} r(m+n) &= rm + rn, \\ (r+s)m &= rm + sm, \\ (rs)m &= r(sm), \\ 1m &= m. \end{aligned}\right. \quad \text{for any } r, s \in R, m, n \in M$$

A ***right*** *R-**module** is defined by $(r, m) \mapsto mr$ with the axiom $m(rs) = (mr)s$. If R is commutative, left-modules and right-modules coincide. Otherwise, one can transform using the opposite ring R^{op}.*

Consider M an R-module, then $\forall r \in R$, $\varphi_r : M \to M$ by $\varphi_r(m) = rm$ is an abelian group homomorphism. Let

$$\mathrm{End}(M) = \{\varphi : M \to M \text{ is a abelian group homomorphism }\}$$

Then $\mathrm{End}(M)$ is a ring with composition as the multiplication.

Theorem 1.1.2. *Let M be an abelian group. Then M is an R-module if and only if there is a ring homomorphism $R \to \mathrm{End}(M)$.*

Proof. ($\Rightarrow$) Suppose M is an R-module. Define $\rho : R \to \mathrm{End}(M)$ by

$$\rho(r)(m) = rm, \quad \forall r \in R, m \in M.$$

For each $r \in R$, the map $\rho(r) : M \to M$ is a group homomorphism since $r(m_1 + m_2) = rm_1 + rm_2$. Moreover, ρ is a ring homomorphism:

- $\rho(r+s)(m) = (r+s)m = rm + sm = \rho(r)(m) + \rho(s)(m)$, so $\rho(r+s) = \rho(r) + \rho(s)$.
- $\rho(rs)(m) = (rs)m = r(sm) = \rho(r)(\rho(s)(m))$, so $\rho(rs) = \rho(r) \circ \rho(s)$.
- $\rho(1)(m) = 1m = m$, so $\rho(1) = \mathrm{id}_M$.

($\Leftarrow$) Suppose $\rho : R \to \mathrm{End}(M)$ is a ring homomorphism. Define the scalar multiplication $R \times M \to M$ by

$$(r, m) \mapsto \rho(r)(m).$$

We verify the module axioms:

- $r(m_1 + m_2) = \rho(r)(m_1 + m_2) = \rho(r)(m_1) + \rho(r)(m_2) = rm_1 + rm_2$.
- $(r+s)m = \rho(r+s)(m) = (\rho(r) + \rho(s))(m) = \rho(r)(m) + \rho(s)(m) = rm + sm$.
- $(rs)m = \rho(rs)(m) = (\rho(r) \circ \rho(s))(m) = \rho(r)(\rho(s)(m)) = r(sm)$.
- $1m = \rho(1)(m) = \mathrm{id}_M(m) = m$.

Hence M is an R-module. □

Example 1.1.3. *(1) A vector space is a module over a field.*
(2) An abelian group is a module over the ring of integers $\mathbb{Z}$.

$$\mathbb{Z} \times M \to M, \quad (n, m) \mapsto n \cdot m := m + m + \cdots + m \ (n \text{ times})$$

(3) An ideal I of a ring R is a module over R.
(4) Let M be an abelian group. Then M is an $\mathrm{End}(M)$-module via

$$\mathrm{End}(M) \times M \to M, \quad (\varphi, m) \mapsto \varphi(m).$$

(5) Let V be a vector space over a field K, and let $A : V \to V$ be a linear transformation. Then V is a $K[x]$-module via

$$K[x] \times V \to V, \quad (f(x), v) \mapsto f(A)v.$$

(6) If $|M| = 1$, then M is called the zero module.
(7) Let R be a ring with 1 and $A \in M_n(R)$. The solution of $AX = 0$ where $X \in R^n$ is a right R-module.

Definition 1.1.4 (R-submodule)**.** *Let R be a ring (with 1) and let M be a left R-module. A subset $N \subseteq M$ is called an R-submodule of M if:*

1. $0 \in N$;
2. $n_1, n_2 \in N \implies n_1 + n_2 \in N$;

3. $n \in N \implies -n \in N$;

4. $r \in R,\ n \in N \implies rn \in N$.

Equivalently, N is a subgroup of the abelian group $(M, +)$ that is closed under the R-action.

Example 1.1.5. *(1) Let V be a vector space over a field K, then any subspace of V is a k-submodule of V.*
(2) Let G be an abelian group, then any subgroup of G is a $\mathbb{Z}$-submodule of G.
(3) Let R be a ring, ${}_RR$ is a left (regular) R-module. Then any left ideal of R is a R-submodule of ${}_RR$.
(4) Let G be an abelian group, which is an $\mathrm{End}(G)$-module. Then any full invariant subgroup of G is a $\mathrm{End}(G)$-submodule of G.

Question: Are submodules of a finitely generated R-module finitely generated?
Answer: No, not in general. This property holds if and only if the ring R is Noetherian.

Example 1.1.6 (Submodule of a f.g. module that is not f.g.)**.** *Let k be a field and consider the polynomial ring $R = k[x_1, x_2, x_3, \dots]$ in infinitely many variables.*

- *The ring R itself can be viewed as a left R-module, which we denote by $M = R$. M is finitely generated, for instance by the single element $1 \in R$.*
- *Consider the ideal $I = \langle x_1, x_2, x_3, \dots \rangle$ generated by all the variables. This ideal I is a submodule of $M = R$.*
- *This submodule I is not finitely generated. To see this, suppose for contradiction that I is finitely generated by a set of polynomials $\{f_1, \dots, f_n\}$. Let N be the largest index of any variable appearing in any of the polynomials $f_1, \dots, f_n$. Then any element in the ideal generated by $\{f_1, \dots, f_n\}$ must be a polynomial in variables $x_1, \dots, x_N$. However, the polynomial x_{N+1} is in I but cannot be expressed as a combination of $f_1, \dots, f_n$. This is a contradiction.*

Therefore, the ideal I is a submodule of the finitely generated R-module R which is not itself finitely generated. This shows that the ring R is not Noetherian.

Definition 1.1.7. *Let M be an R-module, $M_i, i \in I$ are R-submodules of M. Then*

$$\bigcap_{i \in I} M_i$$

is an R-submodule of M.

$$\sum_{i \in I} = \{ \sum_{i \in I\ finite} m_i | m_i \in M_i \}$$

is an R-submodule of M.

Definition 1.1.8. *Let M be an R-module, $S \subseteq M$ a subset.*

$$< S >= \{ \sum_{finite} a_i s_i | a_i \in R, s_i \in S\}$$

or

$$< S >= \bigcap_{N is\ an\ R-submodule\ of\ M,\ S \subseteq N} N$$

is an R-submodule of M, called the R-submodule of M generated by S.
If $M =< S >$, then we say M is generated by S. In particular, if $|S| = 1$ then we call M a cyclic module. If $|S| < \infty$ then we call M a finite generated module.

Definition 1.1.9. *Let R be a ring and let M be a left R-module. We say that M is a* Noetherian module *if every ascending chain of submodules*

$$M_1 \subseteq M_2 \subseteq M_3 \subseteq \cdots$$

stabilizes; that is, there exists $n \geq 1$ such that

$$M_n = M_{n+1} = M_{n+2} = \cdots .$$

Equivalently, M is Noetherian if every submodule of M is finitely generated.

1.2 Quotient Modules and Homomorphism Theorems

Definition 1.2.1. *Let R be a ring, let M be a left R-module, and let N be a submodule of M. The* quotient module *(or* factor module*) of M by N is the quotient abelian group*

$$M/N = \{ x + N \mid x \in M \},$$

with scalar multiplication defined by

$$r(x + N) = rx + N \qquad (r \in R,\ x \in M).$$

This is well-defined, and with this operation M/N becomes a left R-module.

Example 1.2.2. *(1) Let R be a ring and an R-module, I be an ideal of R, then R/I is a quotient R-module of R by I.*
(2) Consider the ring $\mathbb{Z}$ and the module $\mathbb{Z}$ over $\mathbb{Z}$, then for $n \in \mathbb{Z}$, $n\mathbb{Z}$ is a submodule of $\mathbb{Z}$, and $\mathbb{Z}/n\mathbb{Z}$ is a quotient $\mathbb{Z}$-module of $\mathbb{Z}$ by $n\mathbb{Z}$. Furthermore, $\mathbb{Z}/n\mathbb{Z}$ is a cyclic module.

Definition 1.2.3. *Let R be a ring, and let M and N be left R-modules. A group homomorphism*

$$f : M \to N$$

*is called an R-*module homomorphism *(or R-*linear map*) if for all $x, y \in M$ and all $r \in R$, one has*

$$f(rx) = rf(x) \quad \forall r \in R, x \in M.$$

Definition 1.2.4. *Let f be an R-module homomorphism. The* kernel *of f is the set*

$$\ker(f) = \{x \in M \mid f(x) = 0\}.$$

$\ker(f)$ *is a submodule of M. The* image *of f is the set*

$$\mathrm{im}(f) = \{f(x) \in N \mid x \in M\}.$$

$\mathrm{im}(f)$ *is a submodule of N.*
If $\ker(f) = 0$, then f is injective. If $\mathrm{im}(f) = N$, then f is surjective. If $\ker(f) = 0$ and $\mathrm{im}(f) = N$, then f is bijective.
A bijective R-module homomorphism is called an isomorphism.

Example 1.2.5. *Let R be a ring and M, N be an R-modules. Then $\mathrm{Hom}_R(M, N)$ is an R-module when R is commutative.*

Theorem 1.2.6. *If $f : M \to N$ is an R-module homomorphism, then we have a canonical R-module isomorphism:*

$$M/\ker(f) \to \mathrm{im}(f)$$

Theorem 1.2.7. *Let N be an R-submodule of M, $\pi_N : M \to M/N$ be a quotient map. Then π_N induces a natural bijection:*

$$\{R - \textit{submodules of } M \textit{ containing } N\} \to \{R - \textit{submodules of } M/N\}$$

Theorem 1.2.8. *If $N \subseteq H$ are two R-submodules of M, then*

$$M/H \cong M/N/(H/N),$$

Theorem 1.2.9. *If N, H are R-submodules of M, then*

$$\frac{H + N}{N} \cong \frac{H}{H \cap N}$$

Proposition 1.2.10. *Let M be an R-module, N a submodule of M, M' an R-module, and $\varphi : M \to M'$ a morphism such that $N \subseteq \ker(\varphi)$. Then there exists a unique morphism*

$$\varphi_N : M/N \to M'$$

such that the following diagram commutes. Moreover, $\ker(\varphi_N) = \ker\varphi/N$.

$$\begin{tikzcd} M \arrow[r, "\varphi"] \arrow[d, "\pi_N"'] & M' \\ M/N \arrow[ur, dashed, "\varphi_N"'] & \end{tikzcd}$$

Remark 1.2.11. *The pair $(M/N, \pi_N)$ is unique in the following sense: if (S, θ) is a pair such that*

- *S is an R-module,*
- *$\theta : M \to S$ is a surjective morphism satisfying the property of $(M/N, \pi_N)$ described in the above proposition,*

then there exists a unique isomorphism $\psi : M/N \to S$ such that the following diagram commutes.

$$\begin{tikzcd} M \arrow[r, two heads, "\theta"] \arrow[d, two heads, "\pi_N"'] & S \\ M/N \arrow[ur, dashed, "\exists!\,\psi"'] & \end{tikzcd}$$

Proof. Since (S, θ) satisfies the same universal property as $(M/N, \pi_N)$, we may apply that property to the canonical projection

$$\pi_N : M \to M/N.$$

Because $N \subseteq \ker(\pi_N)$, there exists a unique morphism

$$u : S \to M/N$$

such that

$$u \circ \theta = \pi_N.$$

On the other hand, since $(M/N, \pi_N)$ has the universal property of the quotient by N, and since $N \subseteq \ker(\theta)$, there exists a unique morphism

$$\psi : M/N \to S$$

such that

$$\psi \circ \pi_N = \theta.$$

We now show that ψ and u are inverse to each other.

First, compute
$$(u \circ \psi) \circ \pi_N = u \circ (\psi \circ \pi_N) = u \circ \theta = \pi_N.$$
But the identity morphism $\mathrm{id}_{M/N} : M/N \to M/N$ also satisfies
$$\mathrm{id}_{M/N} \circ \pi_N = \pi_N.$$
By the uniqueness part of the universal property of $(M/N, \pi_N)$, it follows that
$$u \circ \psi = \mathrm{id}_{M/N}.$$
Similarly,
$$(\psi \circ u) \circ \theta = \psi \circ (u \circ \theta) = \psi \circ \pi_N = \theta.$$
But the identity morphism $\mathrm{id}_S : S \to S$ also satisfies
$$\mathrm{id}_S \circ \theta = \theta.$$
By the uniqueness part of the universal property of (S, θ), we obtain
$$\psi \circ u = \mathrm{id}_S.$$
Therefore ψ is an isomorphism, with inverse u.

Finally, the uniqueness of ψ follows directly from the universal property of $(M/N, \pi_N)$: indeed, ψ is the unique morphism
$$\psi : M/N \to S$$
such that
$$\psi \circ \pi_N = \theta.$$
Hence there exists a unique isomorphism
$$\psi : M/N \to S$$
for which the diagram commutes. □

Proposition 1.2.12. *There is a unique isomorphism* $\bar{\varphi} : \operatorname{coim}\varphi \to \operatorname{im}\varphi$ *such that the following diagram commutes:*

$$\begin{tikzcd} M \arrow[r, "\varphi"] \arrow[d, "\pi_{\ker(\varphi)}"'] & M' \\ \operatorname{coim}(\varphi) \arrow[r, "\overline{\varphi}"] & \operatorname{im}(\varphi) \arrow[u, hook] \end{tikzcd}$$

Recall:

$$\begin{aligned}&\varphi : M \to M', \quad R\text{-morphism}\\&\operatorname{coker}(\varphi) = M'/\operatorname{im}(\varphi)\\&\operatorname{coim}(\varphi) = M/\ker\varphi\end{aligned}$$

Definition 1.2.13. *Let M be an R-module and I an ideal of R. Denote by IM the submodule of M generated by all elements am where $a \in I$, $m \in M$.*

$$IM = \left\{ \sum_{\text{finite}} a_i m_i \mid a_i \in I,\ m_i \in M \right\}$$

Proposition 1.2.14. *Let M be an R-module. Then $R \to \operatorname{End}(M)$ is a ring homomorphism.*

1. *$R \to \operatorname{End}(M)$ factors through the natural surjection $R \to R/I$ if and only if $IM = 0$:*

$$\begin{tikzcd} R \arrow[rr, "\rho"] \arrow[dr, twoheadrightarrow, "\pi_I"'] & & \operatorname{End}(M) \\ & R/I \arrow[ur, dashed, "\exists \rho_I"'] & \end{tikzcd}$$

   Where ρ factors through π_I means $\exists \rho_I : R/I \to \operatorname{End}(M)$ such that the diagram above commutes.

2. *The module M/IM is naturally endowed with a structure of R/I-module.*

3. *Let $\varphi : M \to M'$ be an R-homomorphism. Then $\varphi(IM) \subseteq IM'$, so φ induces a morphism of R/I-modules*

$$M/IM \to M'/IM'.$$

4. *If $M = M_1 \oplus \cdots \oplus M_n$, then $M/IM = M_1/IM_1 \oplus \cdots \oplus M_n/IM_n$.*

5. *$R^n/IR^n \cong (R/IR)^n$.*

Proof. **(1)** Suppose first that ρ factors through π_I. Then there exists a ring homomorphism

$$\rho_I : R/I \to \operatorname{End}_{\mathbb{Z}}(M)$$

such that

$$\rho = \rho_I \circ \pi_I.$$

If $a \in I$, then $\pi_I(a) = 0$, so

$$\rho(a) = \rho_I(0) = 0.$$

Therefore $am = 0$ for every $m \in M$. Since IM is generated by elements of the form am with $a \in I$ and $m \in M$, it follows that

$$IM = 0.$$

Conversely, assume that $IM = 0$. Then for every $a \in I$ and every $m \in M$ we have

$$\rho(a)(m) = am = 0.$$

Thus $\rho(a) = 0$, so $I \subseteq \ker \rho$. Hence ρ factors uniquely through the quotient ring R/I: there exists a unique ring homomorphism

$$\rho_I : R/I \to \mathrm{End}_{\mathbb{Z}}(M)$$

such that

$$\rho = \rho_I \circ \pi_I.$$

(2) Define an action of R/I on M/IM by

$$(r + I) \cdot (m + IM) := rm + IM.$$

We must show that this is well-defined.

If $r + I = r' + I$, then $r - r' \in I$, so

$$(r - r')m \in IM,$$

hence

$$rm + IM = r'm + IM.$$

If $m + IM = m' + IM$, then $m - m' \in IM$. Since IM is a submodule of M, we have

$$r(m - m') \in IM,$$

so

$$rm + IM = rm' + IM.$$

Thus the action is well-defined. The module axioms follow immediately from the module axioms on M. Therefore M/IM is naturally an R/I-module.

(3) Let $\varphi : M \to M'$ be an R-module homomorphism. Take any element of IM, say

$$x = \sum_{j=1}^{t} a_j m_j \qquad (a_j \in I,\ m_j \in M).$$

Then

$$\varphi(x) = \sum_{j=1}^{t} \varphi(a_j m_j) = \sum_{j=1}^{t} a_j \varphi(m_j).$$

Since each $a_j \in I$ and each $\varphi(m_j) \in M'$, it follows that

$$a_j\varphi(m_j) \in IM'.$$

Hence $\varphi(x) \in IM'$, and therefore

$$\varphi(IM) \subseteq IM'.$$

Thus we may define

$$\overline{\varphi} : M/IM \to M'/IM', \qquad \overline{\varphi}(m + IM) = \varphi(m) + IM'.$$

This is well-defined: if $m + IM = m' + IM$, then $m - m' \in IM$, so

$$\varphi(m) - \varphi(m') = \varphi(m - m') \in IM',$$

hence

$$\varphi(m) + IM' = \varphi(m') + IM'.$$

Finally, $\overline{\varphi}$ is R/I-linear, since

$$\overline{\varphi}\big((r+I)\cdot(m+IM)\big) = \overline{\varphi}(rm+IM) = \varphi(rm)+IM' = r\varphi(m)+IM' = (r+I)\cdot\overline{\varphi}(m+IM).$$

(4) Assume that

$$M = M_1 \oplus \cdots \oplus M_n.$$

We first show that

$$IM = IM_1 \oplus \cdots \oplus IM_n.$$

If $x \in IM$, then x is a finite sum of elements of the form am with $a \in I$ and $m \in M$. Writing

$$m = m_1 + \cdots + m_n \qquad (m_i \in M_i),$$

we get

$$am = am_1 + \cdots + am_n,$$

and each $am_i \in IM_i$. Hence $x \in IM_1 + \cdots + IM_n$, so

$$IM \subseteq IM_1 + \cdots + IM_n.$$

The reverse inclusion is immediate because each $M_i \subseteq M$, hence each $IM_i \subseteq IM$. Therefore

$$IM = IM_1 + \cdots + IM_n.$$

To see that the sum is direct, suppose

$$x_1 + \cdots + x_n = 0 \qquad \text{with } x_i \in IM_i \subseteq M_i.$$

Since

$$M = M_1 \oplus \cdots \oplus M_n$$

is a direct sum, we must have $x_i = 0$ for all i. Thus

$$IM = IM_1 \oplus \cdots \oplus IM_n.$$

Now define

$$\Psi : M_1/IM_1 \oplus \cdots \oplus M_n/IM_n \to M/IM$$

by

$$\Psi(m_1 + IM_1, \ldots, m_n + IM_n) = (m_1 + \cdots + m_n) + IM.$$

This map is well-defined and R/I-linear. It is surjective because every element of M can be written as $m_1 + \cdots + m_n$. It is injective because

$$(m_1 + \cdots + m_n) + IM = IM$$

means that

$$m_1 + \cdots + m_n \in IM = IM_1 \oplus \cdots \oplus IM_n,$$

so each $m_i \in IM_i$. Hence

$$m_i + IM_i = IM_i$$

for all i, and the kernel is trivial. Therefore Ψ is an isomorphism:

$$M/IM \cong M_1/IM_1 \oplus \cdots \oplus M_n/IM_n.$$

(5) Since

$$R^n = R \oplus \cdots \oplus R$$

(n copies), part (4) yields

$$R^n/IR^n \cong R/IR \oplus \cdots \oplus R/IR.$$

Therefore

$$R^n/IR^n \cong (R/IR)^n.$$

This completes the proof. □

1.3 Direct Sums and Free Modules

Definition 1.3.1. *Direct sum and direct product* $\left(\bigoplus_{i\in I} M_i \text{ \& } \prod_{i\in I} M_i\right)$.
Let $(M_i)_{i\in I}$ *be a family of* R*-modules. The direct product of* $(M_i)_{i\in I}$ *is the module*

$$\prod_{i\in I} M_i = \{\varphi : I \to \cup_{i\in I} M_i | \forall i \in I, \varphi(i) \in M_i\}$$

with the componentwise addition and scalar multiplication. i.e.

$$R \times \prod_{i\in I} M_i \to \prod_{i\in I} M_i, \quad (r,\varphi) \mapsto (r\varphi(i))_{i\in I}$$

The direct sum of $(M_i)_{i\in I}$ *is the module*

$$\bigoplus_{i\in I} M_i = \{\varphi \in \prod_{i\in I} M_i | \sharp\{i\in I|\varphi(i)\neq 0\} < \infty\}$$

Equal when $|I| < \infty$. *In a module category, direct product = product, direct sum = coproduct.*

Theorem 1.3.2. *Suppose that* $M_1,\dots,M_n$ *are* R*-submodules of* M *in which* $M = M_1 + \cdots + M_n$, $I = [n]$. *Then the followings are equivalent:*

1. *The map* $\varphi : \sum_{i\in I} M_i \to M$ *by* $\varphi((m_i)_{i\in I}) = \sum_{i\in I} m_i, m_i \in M_i$ *is an isomorphism.*
2. *Any element of* M *can be uniquely written as a sum of elements of* M_i.
3. 0 *can be uniquely written as a sum of elements of* M_i. *More precisely,* $0 = 0 + \ldots + 0$.
4. *For* $i \in I$, $M_i \cap (\sum_{j\neq i} M_j) = 0$.

Proof. We prove

$$(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (4) \Rightarrow (1).$$

$(1) \Rightarrow (2)$. If φ is an isomorphism, then for every $m \in M$ there exists a unique

$$(m_1,\dots,m_n) \in \sum_{i=1}^{n} M_i$$

such that

$$m = \varphi(m_1,\dots,m_n) = m_1 + \cdots + m_n.$$

Hence every element of M can be written uniquely as a sum of elements of the M_i.

$(2) \Rightarrow (3)$. Apply (2) to the element $0 \in M$.

$(3) \Rightarrow (4)$. Fix $i \in I$, and let

$$x \in M_i \cap \sum_{j\neq i} M_j.$$

Then $x \in M_i$, and there exist $m_j \in M_j$ for $j \neq i$ such that

$$x = \sum_{j\neq i} m_j.$$

Therefore

$$0 = x - \sum_{j \neq i} m_j$$

is a decomposition of 0 as a sum of elements from the submodules $M_1, \ldots, M_n$. By the uniqueness in (3), this decomposition must be the trivial one. In particular,

$$x = 0.$$

Hence

$$M_i \cap \sum_{j \neq i} M_j = 0.$$

(4) $\Rightarrow$ (1). Since

$$M = M_1 + \cdots + M_n,$$

the map φ is surjective. It remains to show that φ is injective.

Suppose that

$$\varphi(m_1, \ldots, m_n) = 0,$$

that is,

$$m_1 + \cdots + m_n = 0 \qquad \text{with } m_i \in M_i.$$

For each $i \in I$, we have

$$m_i = -\sum_{j \neq i} m_j \in \sum_{j \neq i} M_j.$$

Since also $m_i \in M_i$, it follows that

$$m_i \in M_i \cap \sum_{j \neq i} M_j = 0.$$

Thus $m_i = 0$ for all i, so

$$\ker \varphi = 0.$$

Therefore φ is injective, and hence an isomorphism.

This proves that the four conditions are equivalent. □

Definition 1.3.3 (free module)**.** *Let M be an R-module.*

1. *$x_1, \ldots, x_n \in M$ are linearly independent if*

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = 0 \Rightarrow a_1 = a_2 = \cdots = a_n = 0.$$

2. *M is called free if it is generated by a linearly independent subset. This subset is called a basis of M.*

3. *If M is an R-module with a basis $x_1, \ldots, x_n$, then $M = \langle x_1 \rangle \oplus \cdots \oplus \langle x_n \rangle$.*

Example 1.3.4. $R \oplus \cdots \oplus R = R^n$.

Theorem 1.3.5. *Let R be a commutative ring with 1. Let M be a finitely generated free R-module. Then any basis of M has the same cardinality. The number of elements in a basis is called the rank of the free module.*

Proof. Let $B = \{x_i\}_{i\in I}$ be a basis of M. We first show that any basis is finite. Since M is finitely generated, choose generators $y_1, \ldots, y_n$ of M. Each y_k is a finite R–linear combination of elements of B, so there exists a finite subset $I_0 \subset I$ such that $y_1, \ldots, y_n \in \sum_{i\in I_0} Rx_i$. Hence

$$M = \sum_{k=1}^{n} Ry_k \subseteq \sum_{i\in I_0} Rx_i \subseteq M,$$

so $M = \sum_{i\in I_0} Rx_i$. If $I \setminus I_0 \neq \varnothing$, pick $j \in I \setminus I_0$. Then $x_j \in M = \sum_{i\in I_0} Rx_i$, contradicting that B is linearly independent. Thus $I = I_0$ is finite.

Now let $x_1, \ldots, x_r$ and $y_1, \ldots, y_s$ be two bases of M. Let $J \subset R$ be a maximal ideal and set $F := R/J$, a field. Then M/JM is naturally an F-vector space. Write $\overline{m}$ for the class of $m \in M$ in M/JM, and $\overline{a}$ for the class of $a \in R$ in F. We claim that $\overline{x_1}, \ldots, \overline{x_r}$ form an F–basis of M/JM.

Spanning. For any $m \in M$, write $m = \sum_{i=1}^r a_i x_i$. Then $\overline{m} = \sum_{i=1}^r \overline{a_i}\,\overline{x_i}$, so $\overline{x_1}, \ldots, \overline{x_r}$ span M/JM.

Linear independence. Suppose $\sum_{i=1}^r \overline{a_i}\,\overline{x_i} = 0$ in M/JM. Then $\sum_{i=1}^r a_i x_i \in JM$, so there exist $a'_1, \ldots, a'_r \in J$ such that $\sum_{i=1}^r a_i x_i = \sum_{i=1}^r a'_i x_i$. Hence $\sum_{i=1}^r (a_i - a'_i)x_i = 0$. By linear independence of $x_1, \ldots, x_r$, we get $a_i - a'_i = 0$ for all i, so $a_i \in J$, thus $\overline{a_i} = 0$ for all i. Therefore $\overline{x_1}, \ldots, \overline{x_r}$ are F-linearly independent, and form an F–basis of M/JM.

By the same argument, $\overline{y_1}, \ldots, \overline{y_s}$ also form an F–basis of M/JM. Hence $r = \dim_F(M/JM) = s$. Therefore any two bases of M have the same cardinality. $\square$

Chapter 2

Lecture 2

Definition 2.0.1. *Let M be an R-module, $m \in M$, $Rm =< m >= \{rm \mid r \in R\}$. Define the annihilator of m in R as:*

$$\operatorname{Ann}_R(m) = \{r \in R \mid rm = 0\}.$$

It is a left ideal of R.

Then we have $R/\operatorname{Ann}_R(m) \cong Rm$ is an isomorphism. Which shows that all cyclic modules are isomorphic to a quotient of R.

Definition 2.0.2. *An element $m \in M$ is called*

- ***torsion-free*** *if* $\operatorname{Ann}_R(m) = 0$.
- ***torsion*** *if* $\operatorname{Ann}_R(m) \neq 0$.

A module is called torsion-free if all its elements are torsion-free.

Definition 2.0.3. *The torsion part of M is defined as:*

$$\operatorname{Tor}(M) = \{m \in M \mid \operatorname{Ann}_R(m) \neq 0\}.$$

Lemma 2.0.4. *If R is an integral domain, then the torsion submodule of M is a submodule of M.*

Proof. Let $m, n \in \operatorname{Tor}(M)$, then since by definition $\operatorname{Ann}_R(m) \neq 0$ and $\operatorname{Ann}_R(n) \neq 0$, we have $\lambda m = \mu n$ for some $\lambda, \mu \neq 0 \in R$. Then $\lambda m = \mu n = 0$. Furthermore, $(\lambda\mu)(m + n) = 0$. Since R is an integral domain, we have $\lambda\mu \neq 0$, hence $m + n \in \operatorname{Tor}(M)$.

Therefore $\operatorname{Tor}(M)$ is a submodule of M. □

Example 2.0.5. *(1) $0 \in M$ is torsion. (2) Let R be a commutative ring with 1. Then R is an integral domain if and only if ${}_RR$ is torsion-free. In this case* $\operatorname{Tor}({}_RR) = 0$,

$\operatorname{Tor}(R^n) = 0$ *for all* $n \geq 1$. *(3) If* I *is a nonzero ideal of* R, *then* $\operatorname{Tor}(R/I) = R/I$ *where* R/I *is regarded as an* R*-module. If we regard* R/I *as an* R/I*-module, then it is torsion-free if and only if* I *is a prime ideal. (4) More generally, let* I *be any non-zero ideal of* R, *and* M *be an* R*-module. Then* $\operatorname{Tor}(M/IM) = M/IM$ *where* M/IM *is regarded as an* R*-module.*

Chapter 3

Homework

3.1 Homework week 1

Exercise 3.1.1. *Let R be a ring, and let M be an R-module. Suppose that S is a ring and*

$$\theta : S \to R$$

is a ring homomorphism satisfying $\theta(1_S) = 1_R$. Let S act on M by

$$sx = \theta(s)x,$$

where $x \in M$ and $s \in S$. Prove that M is an S-module.

Proof. This action is well-defined because $\forall s \in S$, $\theta(s) \in R$ and $\theta(s)x \in M$ for all $x \in M$. We need to verify the module axioms, for all $s, t \in S$ and $x, y \in M$:

- $s(x + y) = \theta(s)(x + y) = \theta(s)x + \theta(s)y = sx + sy$
- $(s + t)x = \theta(s + t)x = \theta(s)x + \theta(t)x = sx + tx$
- $(st)x = \theta(st)x = \theta(s)\theta(t)x = \theta(s)(\theta(t)x) = s(tx)$.
- $1_S x = \theta(1_S)x = 1_R x = x$

Hence M is an S-module. $\square$

Exercise 3.1.2. *Let R be a ring, and let M and N be two right R-modules. Let $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ be the set of group homomorphisms from M to N (as abelian groups). The action of R on $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ is given as follows: for any $a \in R$ and $\varphi \in \mathrm{Hom}_{\mathbb{Z}}(M, N)$, define*

$$(\varphi a)(x) = \varphi(xa), \qquad \forall x \in M.$$

Prove that $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ is a left R-module.

Proof. This action is well-defined. $\forall r \in R$, $\varphi r \in \mathrm{Hom}_{\mathbb{Z}}(M, N)$ for all $\varphi \in \mathrm{Hom}_{\mathbb{Z}}(M, N)$ as $(\varphi r)(x) = \varphi(xr) \in N$ for all $x \in M$. We need to verify the module axioms, for all $r_1, r_2 \in R$ and $\varphi, \psi \in \mathrm{Hom}_{\mathbb{Z}}(M, N)$, $\forall x \in M$:

- $((\varphi + \psi)r)(x) = (\varphi + \psi)(xr) = \varphi(xr) + \psi(xr) = (\varphi r)(x) + (\psi r)(x)$
- $(\varphi(r_1+r_2))(x) = \varphi(x(r_1+r_2)) = \varphi(xr_1+xr_2) = \varphi(xr_1)+\varphi(xr_2) = (\varphi r_1)(x)+(\varphi r_2)(x)$
- $((\varphi r_1)r_2)(x) = (\varphi r_2)(xr_1) = \varphi(xr_2r_1) = (\varphi r_2r_1)(x)$
- $(\varphi 1_R)(x) = \varphi(x1_R) = \varphi(x)$

Hence $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ is a left R-module. □

Exercise 3.1.3. *Let R be a ring, and let M be an R-module. Denote*

$$\mathrm{Ann}_R M = \{r \in R \mid rm = 0, \ \forall m \in M\}.$$

We call M a faithful R-module if $\mathrm{Ann}_R M = \{0\}$. Define the action of the quotient ring $R/\mathrm{Ann}_R M$ on M by

$$\overline{r}\,x = rx,$$

where $x \in M$, $r \in R$, and $\overline{r} = r + \mathrm{Ann}_R M$. Prove that M is a faithful $R/\mathrm{Ann}_R M$-module.

Proof. First we show that the action is well-defined. Suppose

$$\bar{r} = \bar{r}' \in R/\mathrm{Ann}_R M.$$

Then $r - r' \in \mathrm{Ann}_R M$, so for every $x \in M$,

$$(r - r')x = 0.$$

Hence $rx = r'x$, so $\bar{r}\,x$ does not depend on the choice of representative.

The module axioms follow immediately from those of the R-module structure on M.

Now we prove faithfulness. Suppose $\bar{r} \in R/\mathrm{Ann}_R M$ acts trivially on M, that is,

$$\bar{r}\,x = 0 \qquad \forall x \in M.$$

By definition, this means

$$rx = 0 \qquad \forall x \in M.$$

Therefore $r \in \mathrm{Ann}_R M$, so $\bar{r} = \bar{0}$ in the quotient ring. Hence

$$\mathrm{Ann}_{R/\mathrm{Ann}_R M}(M) = \{\bar{0}\},$$

and M is faithful as an $R/\mathrm{Ann}_R M$-module. □

Exercise 3.1.4. *Let $\varphi : M \to M$ be a homomorphism of R-modules such that*

$$\varphi\varphi = \varphi.$$

Prove that

$$M = \ker\varphi \oplus \mathrm{im}\,\varphi.$$

Proof. Let $x \in M$. Then $x = \varphi(x) + (x - \varphi(x))$. Where $\varphi(x) \in \operatorname{im}\varphi$ and $x - \varphi(x) \in \ker\varphi$. Hence $M = \operatorname{im}\varphi + \ker\varphi$. We only need to show that $\ker\varphi \cap \operatorname{im}\varphi = \{0\}$. Suppose $x \in \ker\varphi \cap \operatorname{im}\varphi$. Then $x = \varphi(y)$ for some $y \in M$ and $\varphi(x) = 0$. Then $\varphi(x) = \varphi\varphi(y) = \varphi(y) = x = 0$, so $x = 0$. Hence $\ker\varphi \cap \operatorname{im}\varphi = \{0\}$. □

Exercise 3.1.5. *Show that any R-module is a homomorphism image of some free R-module.*

Proof. Let M be a left R-module. Consider the free R-module

$$F = \bigoplus_{x \in M} Re_x$$

with basis $\{e_x\}_{x\in M}$ indexed by the underlying set of M. Define an R-module homomorphism

$$\pi : F \to M$$

by

$$\pi(e_x) = x \qquad (x \in M),$$

and extend R-linearly. Thus

$$\pi\Big(\sum_{i=1}^{n} r_i e_{x_i}\Big) = \sum_{i=1}^{n} r_i x_i.$$

This is well-defined and R-linear.

For any $m \in M$, we have

$$\pi(e_m) = m,$$

so π is surjective. Therefore M is a homomorphic image of the free module F. Equivalently,

$$M \cong F/\ker\pi.$$

□

Exercise 3.1.6. *Let $\mathbb{Q}$ be the field of rational numbers. Is the additive group $(\mathbb{Q}, +)$ a free $\mathbb{Z}$-module?*

Proof. Claim: any two non-zero element of $\mathbb{Q}$ are linearly dependent over $\mathbb{Z}$. Hence if $\mathbb{Q}$ is a free $\mathbb{Z}$-module, then it must have a basis of cardinality 1.

Proof of claim: let $0 \neq x = \frac{a}{b}$ and $0 \neq y = \frac{c}{d}$ be two elements of $\mathbb{Q}$. Then $dcx - aby = 0$. If $a = 0$ or $c = 0$, then $x = 0$ or $y = 0$, which contradicts the assumption. Hence $a, c \neq 0$ which means dc, ab are two non-zero integer coefficients. Hence x and y are linearly dependent over $\mathbb{Z}$. And if $\mathbb{Q}$ is a free $\mathbb{Z}$-module, then it must have a basis of cardinality 1.

But if $\mathbb{Q}$ is a cyclic $\mathbb{Z}$-module, then it has only one generator. Suppose it is $z = \frac{e}{f}$. Then $\mathbb{Q} = \{n\frac{e}{f} \mid n \in \mathbb{Z}\}$. But this is not possible because we can easily pick $\frac{1}{2f} \notin \mathbb{Q}$, which is absurd.

Therefore $\mathbb{Q}$ is not a free $\mathbb{Z}$-module. □